TailDaam

PAKISTAN FUEL PRICES, WITH RECEIPTS

HACK INSIGHT 2026

TRACK: ENERGY DEMAND

DATASET PROPOSAL

The record behind the price.

Credible sources, provenance and model-use rationale for Pakistan's petrol and diesel record and a Brent volatility model.

62	91,672	3	14
RAW FILES	CLEAN ROWS	PUBLISHERS	QUALITY RULES

91,672 rows are above the 50,000-row gate. No synthetic or re-uploaded data.

PREPARED BY

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21 October 2026 / TailDaam v1.0

THE CLAIM

Every price in the gold layer can be traced to a published row. Unsupported answers refuse.

01 / THE QUESTION

What we are actually modelling

Pakistani petrol and high-speed diesel prices are administered at fortnightly reviews, and the evidence is scattered across PBS reports and global benchmark feeds. Public argument mixes three claims: the notified price, why it moved, and what happens next. TailDaam answers the first with citations, measures the second only descriptively, and does not forecast the administered domestic price.

The primary model targets the risk dimension of the oil import bill: next-month Brent realised volatility. That is where the data has enough history for a defensible baseline comparison.

The agent is refusal-first. A scope gate, retrieval confidence floor, time/product/city filters and a number verifier make unsupported answers a code path, not a prompt request.

PRIMARY TARGET

Next-month Brent realised volatility

BASELINE

Persistence (this month's RV)

RESULT

-7.27% MAE

16.5102 vs 17.8044 on 93 held-out months

NOT CLAIMED

A domestic retail-price forecast or a causal pass-through estimate

02 / SOURCES

Three publishers. 62 files. 91,672 clean rows.

SOURCE	FILES	ROWS	ROLE IN THIS PROJECT	LICENCE
PBS weekly SPI	41	82	Weekly petrol/HSD prices	Government open
PBS monthly annex	9	324	17 cities + national average	Government open
US EIA	11	84,546	Daily spot; Brent is the target	Public domain
World Bank Pink Sheet	1	6,720	Independent energy benchmark	CC BY 4.0
TOTAL	**62 raw files / 91,672 clean rows / 3 independent publishers**			

50,000+

VOLUME GATE

The clean layer is 91,672 rows, not a thin weekly series.

3

INDEPENDENT SOURCES

PBS is the Pakistani anchor; EIA and the World Bank are off the suggested source list.

1

MANIFEST

Every raw file carries a URL, fetch time, SHA-256 and licence.

WHAT IS DELIBERATELY ABSENT

NO SYNTHETIC ROWS

Every observation is published.

NO KAGGLE MIRRORS

Only original publishers are fetched.

NO BACKFILLED WEEKS

Publication gaps stay gaps.

03 / HOW THE DATA IS USED

Training, retrieval, and one descriptive estimate

The model and the agent are deliberately separated. The model is trained on EIA daily Brent only; the agent may quote the PBS, EIA and World Bank gold facts. The Pakistan pass-through is a descriptive small-sample calculation, not a training input and not a causal claim.

MODEL TRAINING

EIA daily Brent Europe spot

Target: next calendar month annualised realised volatility. Features: lagged monthly, quarterly and annual RV. Ridge alpha = 10, pre-registered.

DESCRIPTIVE

PBS weekly + EIA Brent

39 overlapping weeks: correlation 0.6928, slope 0.5821 PKR per USD/bbl. Reported as small-sample evidence; no causal claim.

RAG EVIDENCE

2,417 gold facts

PBS weekly/monthly, EIA and World Bank rows with citation tags and row IDs. The agent may only quote these rows.

04 / PROVENANCE

From publisher file to quoted row

Every transfer is recorded. A gold answer carries the row IDs it came from; those IDs resolve back through the clean table to a raw file and its SHA-256 in data/manifest.csv.

01 **ACQUIRE** original URL > 02 **CLEAN** typed rows > 03 **QUALITY** FIX / FLAG > 04 **GOLD** row IDs > 05 **AGENT** quote only

Manifest fields: file, source, publisher, url, query, fetch_time_utc, sha256, licence, bytes. No Kaggle mirror, no blog scrape, no re-uploaded dataset.

ONE ANSWER, TRACED

What is the current petrol price in Pakistan?

396.06 PKR/L / week ended 2026-10-15 / PBS-SPI / pbs:2026-10-15:petrol_super

The number is quoted from the row, not generated by a language model.

05 / QUALITY AND LIMITS

What is repaired, what is kept, what is refused

QUALITY STATUS

WARN

14 rules / 1 fix / 4 flags / 0 fails

Warnings are kept and shown. A WARN is not hidden behind a PASS.

FIX	Dropped 1 non-positive EIA price; no value invented.
FLAG	20 EIA daily jumps kept: energy markets can genuinely gap.
FLAG	9 Apr 2026 HSD jump 522.29 PKR/L kept and annotated.
FLAG	Two PBS cadence gaps > 28 days; not filled.

06 / LIMITS

The constraints are part of the record

Short history	PBS serves 41 weekly reports; publication gaps stay gaps.
Administered prices	Domestic prices move at policy reviews, not continuously.
City history	PBS city annexes cover Jan-Sep 2026; 2025 editions are not in this build.
Global target	The model forecasts Brent volatility, not the domestic retail level.
Lexical fallback	Without a Mistral key, retrieval is TF-IDF; refusals are preferred to guesses.

07 / REPRODUCIBILITY

One command rebuilds the record

Every layer is deterministic and committed. The raw files stay with their manifest; the clean layer, gold tables, model, metrics and citation registry are regenerated from them.

`make data`	acquire -> clean -> quality -> gold
`make model`	refit baseline and pre-registered challenger
`make test`	contract, refusal and verification tests
`make all`	data, model, index, tests, Summary.pdf, this PDF

DECLARATION

All 91,672 clean rows come from the original PBS, EIA and World Bank publishers. No synthetic, interpolated or AI-generated observation is used for training or evaluation. The quality warnings and the missed pipeline window are disclosed, not backfilled.